

Vision based real time pavement degradation detection

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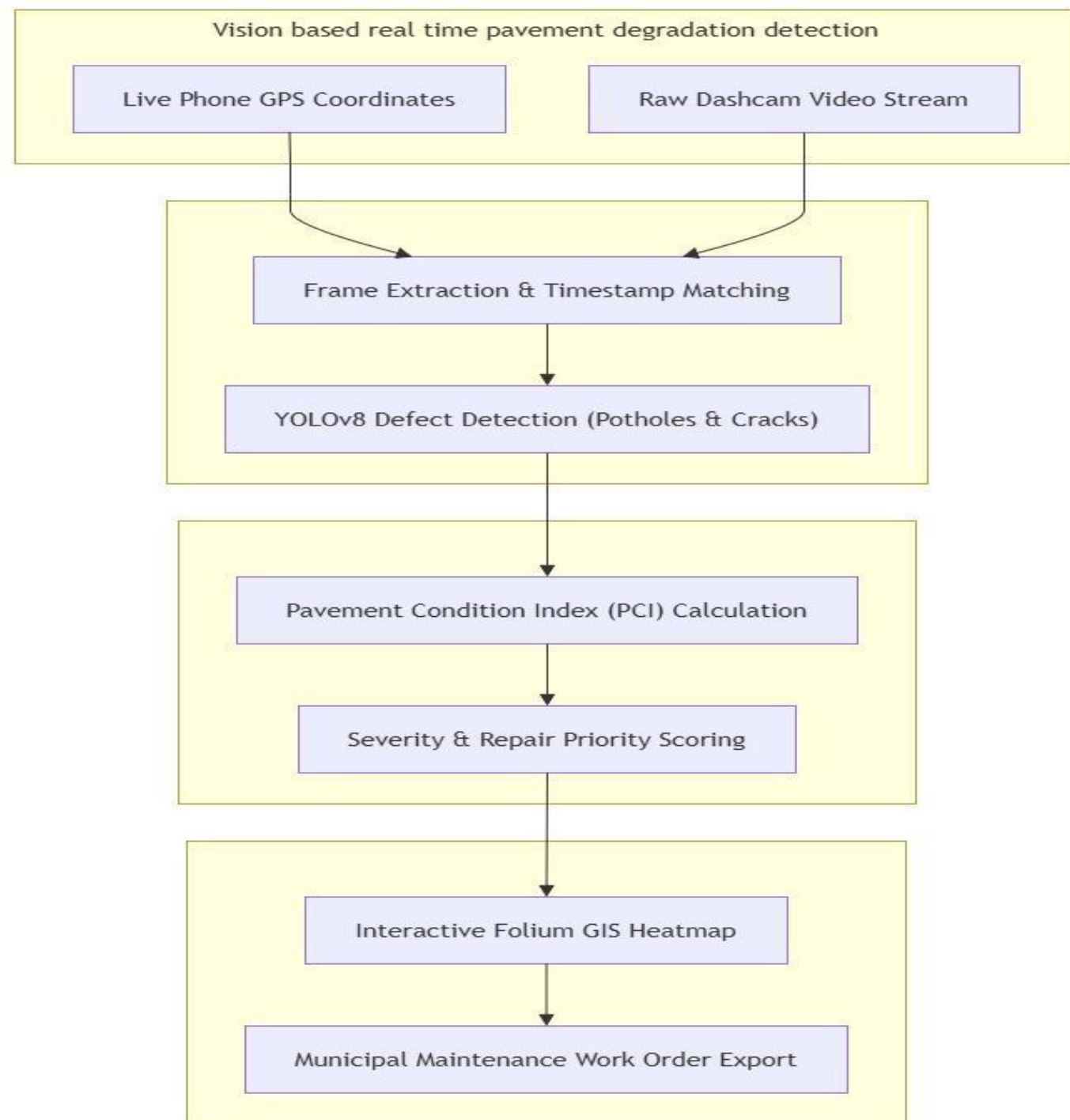
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Problem Statement

- **Manual & Inefficient Surveys:** Traditional road inspections rely on manual human checks, which are slow, expensive, subjective, and miss critical damage until accidents occur.
- **Lack of Spatial Data:** Raw video or photos alone fail to provide precise, real-time GPS locations, making it difficult to pinpoint exact repair spots.
- **Unorganized Repair Prioritization:** Municipalities lack automated tools to measure actual damage severity (like PCI scores), leading to reactive maintenance instead of data-driven repairs.



Methodology

1. Capture: Stream dashcam video and phone GPS together to match pictures with exact locations.
2. Clean: Resize the video frames so the computer can process them quickly.
3. Detect: Use AI to automatically spot and label potholes and cracks.
4. Grade: Check the size of each defect to mark it as minor, medium, or severe.
5. Score: Give the road a simple health grade from 0 to 100.
6. Map: Show red warning dots on a live map and send repair lists to city workers.

References

- [**RDD2022 Dataset Paper**](#): Defines the standardized dataset and codes (D00, D10, D20, D40) used to identify road damage.
- [**YOLOv8 Architecture**](#): Provides the real-time AI model used for high-speed object detection and bounding box placement.
- [**ASTM D6433 Standard**](#): The civil engineering guideline used to compute the 0–100 Pavement Condition Index score.
- [**Web-GIS Review Paper**](#): Demonstrates how to map AI detection results onto interactive GPS dashboards for city planning.
- [**RDD2022 \(Road Damage Dataset 2022\)**](#): Official Repository & Benchmark Paper
- [**India-Specific Subset \(Potholes / D40\)**](#): India-specific pothole subset of RDD2022 with Pascal VOC XML annotations

Thank you